

# Challenge submission

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## 1 Introduction

This submission uses a tensor-basis gene expression programming (TB-GEP) closure discovered with CFD-driven training. Two correction models were trained in a decoupled way: one for the anisotropy correction,  $b_{ij}^\Delta$ , and one for the turbulent kinetic energy correction,  $k_{\text{deficit}}$ .

## 2 Training Setup

The model was trained on seven cases:

DUCT/AR\_1\_Ret\_180, DUCT/AR\_3\_Ret\_180, DUCT/AR\_10\_Ret\_180,  
PHLL10595/PH-Breuer, CBFS13700/CBFS,  
PHLL29/alpha\_05\_4071\_3036, PHLL29/alpha\_15\_13929\_3036.

The CFD-driven stage used the customized `kOmegaSST` solver. Both correction channels were enabled in the solver, but the discovery was decoupled: when training  $b_{ij}^\Delta$ , the  $k_{\text{deficit}}$  field was read from the file obtained using  $k$ -corrective frozen method; when training  $k_{\text{deficit}}$ , the  $b_{ij}^\Delta$  field was read from the file obtained using  $k$ -corrective frozen method too.

## 3 Feature Construction

The tensor bases were generated from two nondimensionalizations of the mean strain-rate and rotation-rate tensors. The first uses a floored local strain scale,

$$\tau_s = \frac{1}{\max(\|\nabla \mathbf{U}\|_F, 10^{-5})}.$$

The second uses an effective SST-consistent omega scale,

$$\tau_\omega = \frac{1}{\max(F_2|S|/0.31, \omega, \omega_{\min})}, \quad |S| = \sqrt{2S_{ij}S_{ij}}.$$

The training used the combined feature family from both timescales.

### 3.1 Notation and Physical Interpretation

The following tables define only the tensorial bases, invariants, dimensional features, and nondimensional features retained by the mutual-information selection step. The suffixes `_s` and `_omega` indicate whether the feature was built from  $\tau_s$  or  $\tau_\omega$ .

| Notation | Expression | Physical interpretation |
|----------|------------|-------------------------|
|----------|------------|-------------------------|

|                |                                                          |                                                                             |
|----------------|----------------------------------------------------------|-----------------------------------------------------------------------------|
| $T_\chi^{(2)}$ | $S_\chi \Omega_\chi - \Omega_\chi S_\chi$                | Nonlinear strain-rotation interaction.                                      |
| $T_\chi^{(3)}$ | $S_\chi^2 - \frac{1}{3} I \text{tr}(S_\chi^2)$           | Deviatoric self-interaction of strain.                                      |
| $T_\chi^{(4)}$ | $\Omega_\chi^2 - \frac{1}{3} I \text{tr}(\Omega_\chi^2)$ | Deviatoric self-interaction of rotation.                                    |
| $S2_\chi$      | $\text{tr}(S_\chi^2)$                                    | Second invariant of nondimensional strain.                                  |
| $W2_\chi$      | $\text{tr}(\Omega_\chi^2)$                               | Second invariant of nondimensional rotation.                                |
| $W2S2_\omega$  | $\text{tr}(\Omega_\omega^2 S_\omega^2)$                  | Mixed strain-rotation invariant.                                            |
| $G_\chi^{(1)}$ | $2k \nabla \mathbf{U} : T_\chi^{(1)}$                    | Dimensional production contribution associated with the first tensor basis. |
| $\epsilon$     | $k\omega$                                                | Turbulent time-scale quantity used by the feature construction.             |

| Notation           | Expression                                                                    | Physical interpretation                                                    |
|--------------------|-------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| $q_Q$              | $\frac{\Omega^2 - S^2}{\max(\Omega^2 + S^2, 10^{-10})}$                       | Q-criterion-style balance between rotation and strain.                     |
| $q_{S\Omega}$      | $\frac{S}{\max(S + \Omega, 10^{-10})}$                                        | Relative local strain magnitude compared with strain plus rotation.        |
| $q_{dp}$           | $\frac{\mathbf{U} \cdot \nabla p}{\max(\mathbf{U} \cdot \nabla p, 10^{-10})}$ | Normalized pressure-gradient projection along the streamline.              |
| $q_T$              | $\frac{kS}{\epsilon + kS}$                                                    | Turbulence-to-mean-strain time-scale ratio.                                |
| $q_\gamma$         | $\frac{\nabla \mathbf{U} k}{\max(\epsilon, 10^{-10})}$                        | Shear-based turbulence parameter.                                          |
| $q_{Re}$           | $2 - \min\left(\frac{\sqrt{k} y_{\text{wall}}}{50\nu}, 2\right)$              | Wall-distance-based turbulence Reynolds-number marker.                     |
| $q_{\tau/k}^{(B)}$ | $\frac{\tau^{(B)}}{\max(k, 10^{-10})}$                                        | Ratio of Boussinesq Reynolds-stress magnitude to turbulent kinetic energy. |
| $q_{D_k/P_k}$      | $\frac{ D_k^{\text{eff}} }{ D_k^{\text{eff}}  +  P_k^{\text{eff}} }$          | Relative magnitude of effective destruction compared with production.      |
| $q_{D_k/C_k}$      | $\frac{ D_k^{\text{eff}} }{ D_k^{\text{eff}}  +  C_k }$                       | Relative magnitude of effective destruction compared with convection.      |
| $q_{P_k/C_k}$      | $\frac{ P_k^{\text{eff}} }{ P_k^{\text{eff}}  +  C_k }$                       | Relative magnitude of effective production compared with convection.       |

Here,  $C_k = \mathbf{U} \cdot \nabla k$  denotes the convection term in the turbulent-kinetic-energy equation,  $P_k$  denotes production,  $D_k$  denotes destruction,  $\tau^{(B)}$  denotes the Reynolds-stress tensor predicted by the baseline Boussi-

nesq closure,  $y_{\text{wall}}$  denotes wall distance, and  $\nu$  denotes molecular kinematic viscosity. The effective RITA terms  $P_k^{\text{eff}}$  and  $D_k^{\text{eff}}$  include the model-propagation contribution when it acts as a local production-like or destruction-like term.

## 4 Training and Selection Metric

The offline and CFD-driven training stages used normalized mean absolute error (NMAE) of velocity  $U$  and TKE  $k$ , so the training metric was aligned with the Challenge Test scoring convention. For the Challenge velocity score in case  $c$ ,

$$E_U^{(c)} = \frac{\frac{1}{N_c} \sum_{i=1}^{N_c} \|\mathbf{U}_i^{\text{pred}} - \mathbf{U}_i^{\text{ref}}\|_2}{\frac{1}{N_c} \sum_{i=1}^{N_c} \|\mathbf{U}_i^{\text{ref}}\|_2}.$$

The final score is the arithmetic mean of  $E_U^{(c)}$  over all test cases.

## 5 Submitted Model

The anisotropy correction model is

$$b_{ij}^{\Delta} = -0.116908104917502 T_{s,ij}^{(2)} + 0.116908104917502 T_{s,ij}^{(3)} + T_{s,ij}^{(4)}.$$

The turbulent kinetic energy correction model is

$$k_{\text{deficit}} = G_{\omega}^{(1)} q_{P_k/C_k} q_{S\Omega}.$$

## 6 Internal Test Score

The internal test sweep score of the submitted model pair is summarized below. Lower values are better.

| Test case           | Submitted model     | Baseline            |
|---------------------|---------------------|---------------------|
| alpha_15_13929_4048 | 0.0431668715        | 0.1320002138        |
| alpha_15_13929_2024 | 0.0998291226        | 0.2048750687        |
| alpha_05_4071_4048  | 0.0743610654        | 0.0461081661        |
| alpha_05_4071_2024  | 0.0960077072        | 0.0718630550        |
| AR_1_Ret_360        | 0.0623051828        | 0.1330176889        |
| AR_3_Ret_360        | 0.0553152679        | 0.1290765849        |
| AR_14_Ret_180       | 0.0526788795        | 0.0682172945        |
| NASA_2DWMH          | 0.0294424690        | 0.0620596613        |
| <b>Overall</b>      | <b>0.0641383207</b> | <b>0.1059022166</b> |